

UNSUPERVISED DEEP LEARNING - ASSIGNMENT 2

PART C: Comparative Analysis

GROUP 78

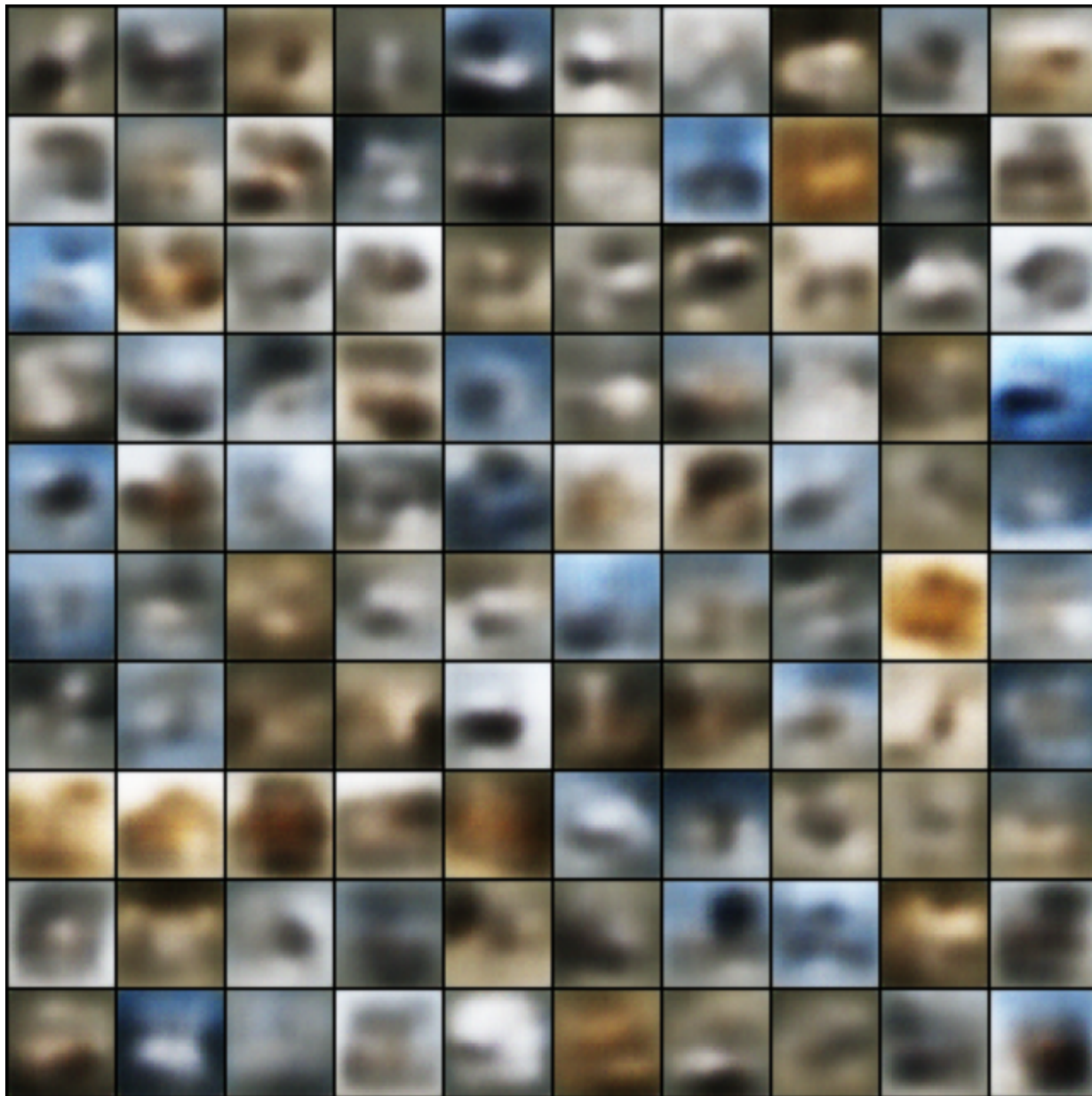
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Includes: 100 generated samples for beta-VAE (beta=1), VQ-VAE (K=256), WGAN and WGAN-GP; evaluation-score comparison; and a discussion of reconstruction error, sampling quality and speed.

Evaluation score comparison

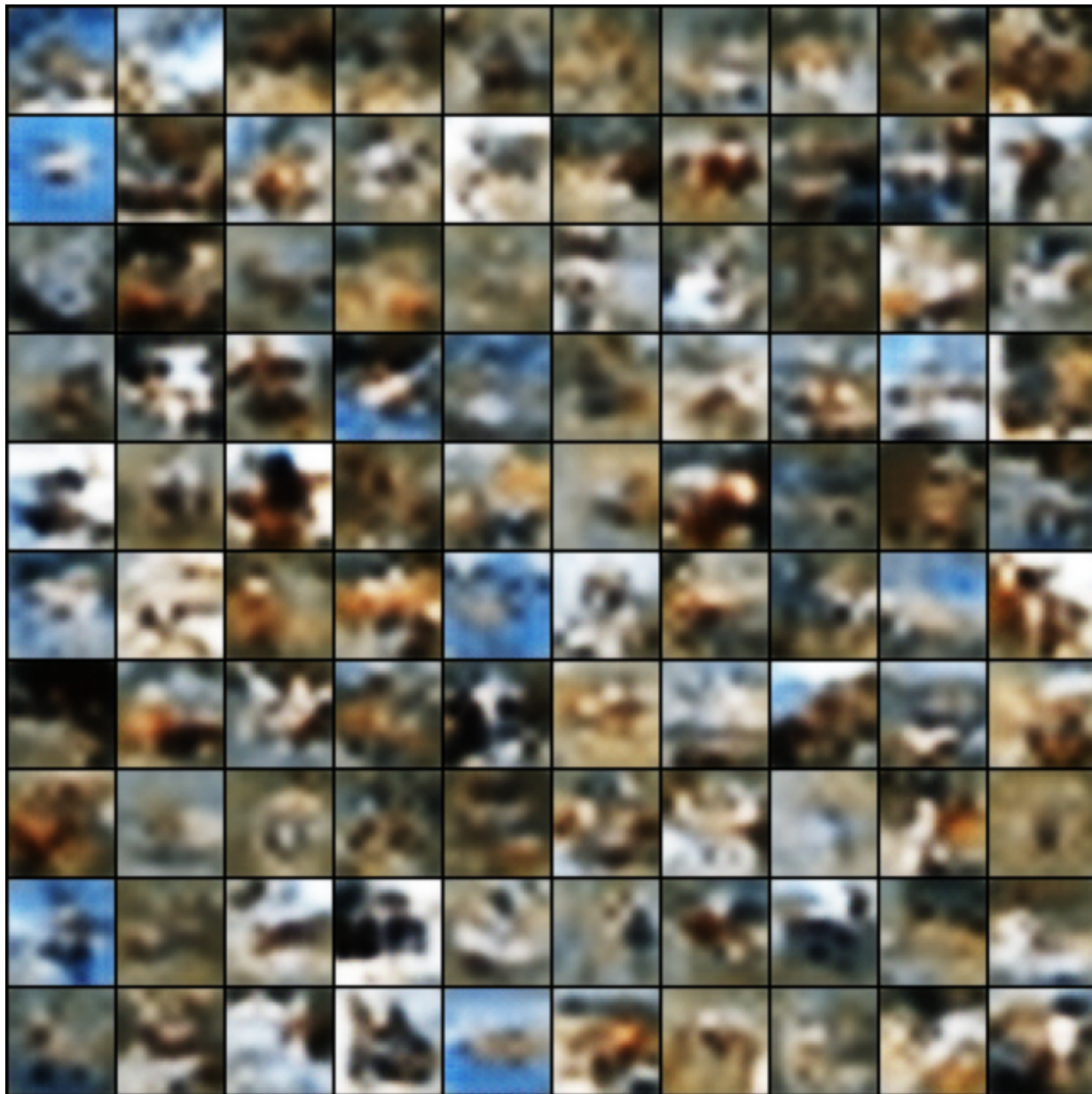
Model	Recon PSNR (dB)	Recon FID	Sample FID	Sampling speed / 100 imgs
beta-VAE (beta=1)	17.52	-	363.83	0.40s (252.4 img/s)
VQ-VAE (K=256) + PixelCNN	20.45	231.63	325.71	42.31s (2.4 img/s)
WGAN	-	-	98.62	not re-benchmarked (see discussion) - single forward pass, no autoregression
WGAN-GP	-	-	80.79	not re-benchmarked (see discussion) - single forward pass, no autoregression

beta-VAE (beta=1) - 100 generated samples



Recon PSNR 17.52 dB | Sample FID 363.83

VQ-VAE (K=256) + PixelCNN - 100 generated samples



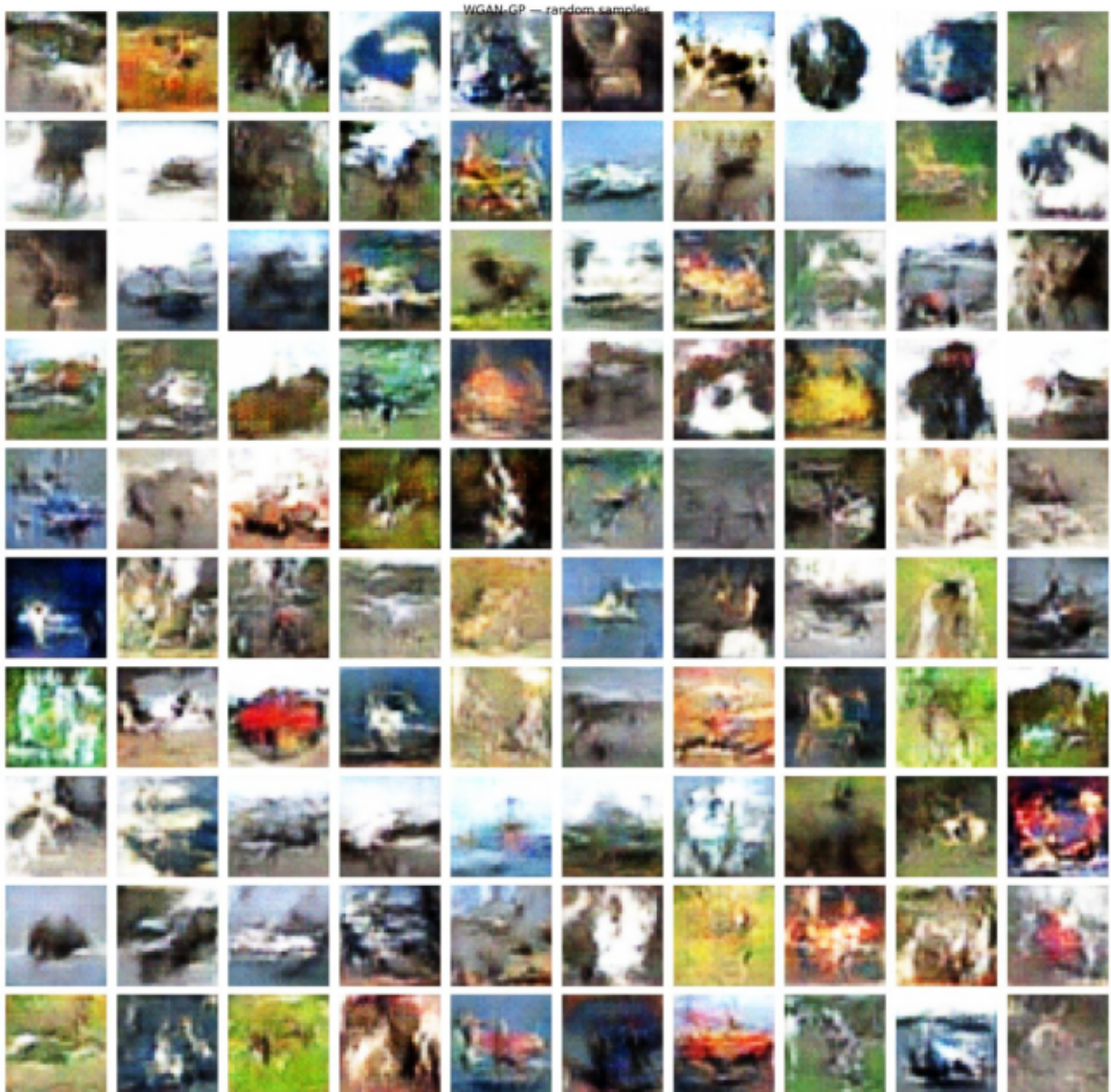
Recon PSNR 20.45 dB | Recon FID 231.63 | Sample FID 325.71

WGAN - 100 generated samples



Sample FID 98.62

WGAN-GP - 100 generated samples



Sample FID 80.79

Comparative discussion

RECONSTRUCTION ERROR

Only the two autoencoder-based models (beta-VAE and VQ-VAE) reconstruct their inputs, so reconstruction error is compared between them via test-set PSNR. VQ-VAE (K=256) reconstructs cleaner than beta-VAE (beta=1): 20.45 dB vs 17.52 dB. This matches the architectural difference - the discrete codebook lets VQ-VAE encode a much higher-capacity representation of each 8x8 grid of local patches than the single 128-dim Gaussian bottleneck used by the VAE, and VQ-VAE has no KL term pulling its latents toward a fixed prior, so it can spend its full capacity on fidelity rather than trading it off against regularization. WGAN and WGAN-GP have no encoder - they only map noise to images - so PSNR is not defined for them.

SAMPLING / PERCEPTUAL QUALITY

Ranked by FID (lower is better): WGAN-GP (80.79) < WGAN (98.62) << VQ-VAE+PixelCNN (325.71) < beta-VAE (363.83). The GANs produce visibly more realistic CIFAR-10-like samples in the grids above - WGAN-GP shows more coherent object silhouettes and fewer flat/noisy patches than WGAN. Both beta-VAE and VQ-VAE+PixelCNN samples are noticeably blurrier and less structured, consistent with the known gap between adversarial and likelihood-based generators: the VAE's pixel-wise MSE objective encourages averaging over plausible outputs (blur), and PixelCNN sampling on top of VQ-VAE compounds a second layer of approximation error on top of the VQ-VAE decoder's own reconstruction error - which is why its sample FID (325.71) is far worse than its reconstruction FID (231.63). Note both Part A models were trained for only 15 epochs on a 15,000-image subset (a deliberate compute-budget tradeoff documented in Part A), so their absolute FID values understate what these architectures can reach with the full dataset and a longer schedule; the relative ranking (GANs > VQ-VAE > VAE) would be expected to hold regardless, though the gap would likely narrow.

SPEED

Sampling speed differs by architecture, not just training budget. beta-VAE and both WGAN variants generate an image with a single decoder/generator forward pass, so 100 samples cost one batched forward pass - measured at 0.40s for the VAE here. WGAN/WGAN-GP were not re-benchmarked in this notebook since their checkpoints were not retained locally between Part B and Part C, but they share the same single-pass architecture as the VAE decoder, so a comparable wall-clock time is expected - certainly nowhere near VQ-VAE+PixelCNN's cost. VQ-VAE+PixelCNN is structurally much slower to sample from: its PixelCNN prior is autoregressive over the 8x8=64-cell latent grid, requiring 64 sequential forward passes through the prior network (one per latent position) before a single batched decode - measured at 42.31s for 100 images here, 107x slower than the VAE on the same GPU and batch size. This sequential-sampling cost is a well-known drawback of PixelCNN-style priors and is the main practical downside of the VQ-VAE+PixelCNN pipeline relative to single-pass generators like VAEs and GANs.

OVERALL

Taking reconstruction error, sample realism (FID) and sampling speed together: WGAN-GP is the strongest generator in this comparison - best FID and fast single-pass sampling - at the cost of providing no encoder/reconstruction pathway. VQ-VAE gives the best reconstruction fidelity and a genuine encoder (useful when compression/reconstruction matters, not just generation), but its PixelCNN-prior sampling is both slower and lower-fidelity than the GANs' output. beta-VAE is the weakest performer on both PSNR and FID among the four at this training budget, though it remains the simplest and cheapest model to train and sample from among the two non-adversarial methods.